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Teacher Manual — Geometry Honors

Florida course 1206320 · Full-year high-school course · 11 modules · Facilitator guide, 2026–2027

🔒 **Facilitator copy.** Contains answer-guidance, rubrics, grading thresholds, and standard codes. **Do not publish to students.**

This manual documents Geometry Honors as it is actually built in Canvas. It is the facilitator's companion for teaching the course to asynchronous On-Demand scholars and, where a section runs live, on the VR campus. Every module count, lesson title, assignment, point value, and standard below was read from the built course; anything that could not be confirmed from the build is marked [VERIFY].

> **How this course differs from the standard Optima template.** Two features assumed by the generic Optima design are **not present** in this build and are therefore not documented as if they existed: there are **no graded discussion forums**, and there are **no Canvas survey-quiz intro/outro module wrappers**. What the course does have is a rich set of auto-checked practice, teacher-graded constructions and proofs, per-lesson VR resources, and module summative assessments. The manual is written to the course that exists.

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1 How this course works

Geometry Honors runs across **11 modules** and four quarters. Semester 1 is **Modules 1–6**; Semester 2 is **Modules 7–11**. Module 9 (Coordinate Geometry) is intentionally split across Q3 and Q4.

A typical lesson

Open Lesson 1.05 (Angles) to see the pattern every lesson follows:

- **Navy header strip** naming Course · Module · Lesson.
- **Content blocks** — definitions, worked examples, and diagrams, with gold "kicker" cards for Essential Questions and key ideas.
- A **VR Lesson Resources** page paired with the lesson — an immersive activity that reinforces the concept (see §2).
- An **Independent Practice** assignment (4–10 problems, ~10 pts, at least two attempts).
- On many lessons, an added **Construction** (compass-and-straightedge) or **proof** — these are the teacher-graded touchpoints.

A typical week (On-Demand)

Segment	What the scholar does	Graded?	Tier
Lesson content	Reads/works 1–2 lesson pages	No	4
VR resource	Completes the paired immersive activity	Participation	4
Independent Practice	4–10 problems, auto-checked, 2 attempts	Yes (auto)	1
Construction / proof	Submits a construction or written proof	Yes (teacher)	3
Mid-Module Check	Checkpoint quiz mid-way through the module	Yes (auto)	1

> Tiers follow `optima-canvas-assignments` (1 = auto-graded practice, 2 = light-touch reflection, 3 = teacher-graded work, 4 = ungraded/participation). This course leans on Tiers 1 and 3; it has no Tier-2 discussions.

Major assessments

Item (per module)	What it asks	Grading	Approx. teacher time
Mid-Module Check(s)	Checkpoint on the first cluster of lessons	Auto	~0 (spot-review)
Summative Review	Full-module review, scored for completion	Completion	~0
Summative Assessment	Full-module test, scored for correctness	Auto + spot-check	~5–10 min/section review
Constructions & proofs	Compass/straightedge work; written proofs	Teacher (rubric)	~4–6 min each

Point totals per module range from ~40 to ~265; the heaviest teacher-grading load is where constructions and proofs cluster — Modules 1, 2, 3, 6, and especially 7 (see §10).

2 Ungraded interactives & VR resources

Two things new facilitators often misread:

- **Ungraded interactive content** inside lesson pages (worked-example reveals, self-checks, diagrams) gives immediate feedback and **costs you nothing to grade** — there is no submission. Treat it as formation-by-practice, not busywork to chase. A scholar skipping it shows up indirectly, in Independent-Practice and Mid-Module-Check performance, not as a missing grade.
- **VR Lesson Resources** pages are paired one-to-one with lessons. They are the course's immersive anchor. Review or assemble captures where the activity produces one; otherwise treat them as participation. Because Optima is moving VR platforms, this manual describes them by *what the scholar does* (build, measure, transform figures in space) rather than by platform feature.

> **No survey-quiz wrappers in this build.** If your school later adds intro/outro reflection wrappers, the high-leverage move is to triage their responses each module for low-confidence clusters. As built, the equivalent early-warning signal here is the **Mid-Module Check** — watch it the same way.

3 AI & VR in this course (student-facing)

AI. The syllabus sets the posture: scholars may use AI to support learning (explain a concept, check understanding, see another worked example) unless an assignment says otherwise, but AI may never do the thinking an assignment asks for — a proof or construction submitted must be genuinely the scholar's own, and any AI use is disclosed briefly. There are no AI-required activities. Speak to it from that stance; do not add new AI requirements.

VR. Every lesson has a paired VR resource. Describe it to students by what they do and produce. If a scholar cannot get into VR for a session, there is an alternative way to take part (see the syllabus) so no one falls behind.

4 Using Ignite AI to support students (teacher-facing)

> **[VERIFY] before relying on specifics.** Ignite AI's exact features and availability vary by Canvas release and by what Optima has enabled. Confirm the current feature set in your Canvas instance; describe capabilities functionally and don't promise a button that isn't turned on.

Used well, the Canvas AI assistant helps one facilitator make a 500:1 load feel personal. In functional terms, use it to:

- **Draft individualized feedback** on a construction or proof, which you then edit and own — faster turnaround, same human judgment.

- **Summarize Mid-Module-Check and Summative analytics** to find the cohort's sticking points fast (e.g., which proof step most scholars miss).
- **Surface scholars who need a nudge** (missed work, declining scores) and draft a warm check-in.
- **Generate practice variants** for a scholar who needs another rep on a Tier-1 skill.
- **Communicate with families** in clear, kind summaries, translating when helpful.

> **AI drafts; the teacher decides.** You always read, edit, and own anything sent to a student or family. AI is never used to surveil, rank, or auto-penalize (see §8).

5 Grading categories & Canvas gradebook setup

> **[VERIFY] weights.** The build stores point values but not official category weights. The scheme below matches the student syllabus defaults (45/20/25/10); confirm against your section's official setup and adjust in one place — the syllabus and gradebook must agree.

Create four assignment groups mapped to the tier system, then set group weights:

Assignment group	Holds	Tier	Method	Suggested weight
Practice & Checks	Independent Practice, Mid-Module Checks	1	Auto	45%
Constructions & Proofs	Construction assignments, proof items (incl. M6 proof quizzes)	3	Rubric	20%
Module Assessments	Summative Reviews (completion) + Summative Assessments (correctness)	1 / 3	Auto + spot	25%
VR & Participation	VR Lesson Resources, lesson completion	4	Completion	10%

Step by step: Gradebook → Settings → *Weight final grade based on assignment groups* → add the four groups above → set weights → set the grading scheme → mark VR/lesson items as **Complete/Incomplete** → set the late/missing policy to match the syllabus (no per-day penalty; end-of-quarter zero; 10% deduction in the final 14 days). Use the **module requirements / progress view** as your at-a-glance completion signal across the ungraded scaffolding.

6 Rubrics for teacher-graded items

Paste each of these into the Canvas item as a **native Canvas Rubric** so it scores in SpeedGrader. Keep them identical across modules so grading stays fast and consistent.

Construction rubric (compass & straightedge)

Criterion	Full	Partial	None
Correct method	Uses valid compass/straightedge steps; no measuring shortcuts	Mostly valid; one shortcut or skipped step	Method invalid or by measurement only
Accuracy of result	Figure is correct within tolerance; arcs/marks preserved	Minor inaccuracy	Result incorrect
Construction marks shown	All arcs and reference marks visible	Some marks missing	No supporting marks

Proof rubric (written / two-column / paragraph)

Criterion	Full	Partial	None
Logical validity	Every step follows; conclusion is established	Mostly valid; one gap	Reasoning breaks down
Justifications	Each step cites a correct postulate/theorem/definition	Some justifications missing or loose	Justifications absent
Precision of notation	Correct symbols, labels, and given/prove	Minor notation errors	Notation unclear

[VERIFY] answer keys. The build routes construction/proof answer keys to unpublished **Teacher Resources** pages (e.g., "Lesson 9.04 Answer Key"). Grade against those keys; never publish them. Confirm each module's Teacher Resources bucket in Canvas before the quarter opens.

7 Assessment types — how to run & grade

Type	What it is	Scoring	Recommended move
Independent Practice	Per-lesson problem set, 4–10 items, 2+ attempts	Auto	Let re-attempts run; scan for the item most miss and address it in the weekly announcement
Mid-Module Check	Checkpoint on early lessons	Auto	Your early-warning signal — flag low scorers for a nudge
Construction	Compass/straightedge submission	Rubric (§6)	Batch-grade with the construction rubric; drop a one-line "next step"
Proof item / M6 proof quizzes	Written proof (free response)	Rubric (§6)	Grade against the Teacher-Resources key; reward valid alternate proofs
Summative Review	Whole-module review	Completion	Completion only — do not hand-grade
Summative Assessment	Whole-module test	Auto + spot-check	Spot-check any free-response sections; use analytics to plan reteach
VR Lesson Resource	Immersive activity paired to a lesson	Participation	Review/assemble captures where produced; otherwise mark participation

8 Coach, not catch

- Feedback names the **next step**, not only the error; lead with what's working in the proof.
- Allow re-attempts on Tier-1 practice — it is a learning tool, not a trap. The two-attempt design exists so scholars review and show gains.
- Use analytics and Ignite AI to **find scholars who need help early**, then reach out — not to build a case.
- On suspected AI misuse or copied proofs, open a conversation: ask the scholar to talk through their reasoning. Treat it as formation, not a gotcha.

9 Weekly announcement scripts

For asynchronous scholars, the weekly announcement is your main presence. Clone this template each week:

> *Scholars — this week you should be in **[Module N, Lessons X–Y]**. Focus: **[the week's big idea]**. Don't miss: **[the due construction/proof or the Mid-Module Check]**. One tip: **[the step most people miss]**. Study hall is **[day/time]** — bring your compass. You've got this. — **[Your name]***

Drafted openers by module (edit freely):

- **M1 Introduction:** "We start with the building blocks — points, lines, planes — and your first constructions. Get your compass out for Lesson 1.02."
- **M2 Logical Reasoning:** "This is where geometry becomes proof. Truth tables first, then your first two-column proofs. The honors proof-by-contradiction is the stretch goal."
- **M3 Congruency:** "SSS, SAS, ASA, AAS, HL — learn when each applies, then prove it. Watch the Mid-Module Check before the summative."
- **M6 Quadrilaterals:** "Two proof quizzes this module (parallelograms, trapezoids). Plan for them — they're teacher-graded."
- **M7 Circles:** "Our heaviest construction module — six of them, including the honors tangent and inscribed-polygon constructions. Pace yourself."

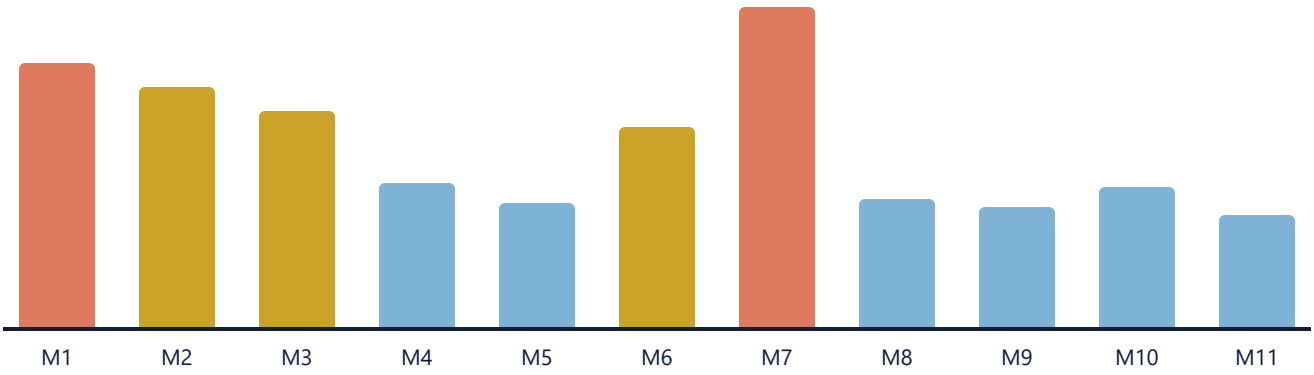
Situational scripts: a mid-module check-in ("If your Mid-Module Check was rocky, come to study hall before you attempt the summative"); a pre-summative script; and a re-engagement nudge for a scholar who's gone quiet.

10 Teacher workflow & workload map

A normal week, in order: scan Mid-Module-Check / practice analytics → batch-grade constructions and proofs with the §6 rubrics → spot-check any summative free-response → post the weekly announcement → answer messages. Tier-1 auto work costs ~0 hours; the real time is in constructions and proofs.

Heavy modules to plan around: M7 (six constructions) is the peak; **M1 (four constructions)**, **M2 & M3 (proof-dense)**, and **M6 (two proof quizzes)** are the other spikes. Measurement, coordinate, and transformation modules (M8–M11) are lighter, mostly auto-graded.

Relative facilitator-grading load by module (estimated from teacher-graded item counts). Blue = light, gold = moderate, coral = heavy.



11 Standards coverage map — where standards are taught

Geometry Honors is aligned to Florida's B.E.S.T. Mathematics standards (CPALMS course 1206320). Below are two views: a quick by-module reference, then the full by-standard map. **These codes are for the teacher copy only** — they do not appear on student pages. Process standards **MA.K12.MTR.1–7**

(mathematical thinking & reasoning) and **ELA.K12.EE.1–6** (expectations for evidence and communication) are embedded throughout every module and are not repeated row by row.

Quick reference — by module

Module	Standards carried
M1 Introduction to Geometry	MA.912.GR.1.1; GR.5.1; GR.5.2
M2 Logical Reasoning	MA.912.LT.4.3; LT.4.8; LT.4.10; GR.1.1 (proofs); GR.5.2 (⊥ bisector)
M3 Triangle Relationships & Congruency	MA.912.GR.1.2; GR.1.3; GR.1.6; GR.2.6; GR.2.7
M4 Similarity & Proportional Relationships	MA.912.GR.1.2; GR.1.6; GR.2.8; GR.2.9
M5 Trigonometry	MA.912.T.1.1; T.1.2; T.1.3; T.1.4
M6 Quadrilaterals	MA.912.GR.1.4; GR.1.5; GR.1.6
M7 Circles	MA.912.GR.5.3; GR.5.4; GR.5.5; GR.6.1; GR.6.2; GR.6.3; GR.6.5
M8 Geometric Measurement	MA.912.GR.4.3; GR.4.4; GR.4.5; GR.4.6; GR.6.4
M9 Coordinate Geometry	MA.912.GR.3.1; GR.3.2; GR.3.3; GR.3.4
M10 Transformations	MA.912.GR.2.1; GR.2.2; GR.2.3; GR.2.4; GR.2.5
M11 Advanced Geometric Applications	MA.912.GR.4.1; GR.4.2; GR.7.2; GR.7.3

Full map — by standard

Standard	Description (abbreviated)	Taught in
MA.912.GR.1.1	Prove/apply theorems about lines and angles	M1 (M2 proofs)
MA.912.GR.1.2	Prove triangle congruence/similarity (SSS, SAS, ASA, AAS, AA, HL)	M3, M4
MA.912.GR.1.3	Prove/apply theorems about triangles	M3
MA.912.GR.1.4	Prove/apply theorems about parallelograms	M6
MA.912.GR.1.5	Prove/apply theorems about trapezoids	M6
MA.912.GR.1.6	Congruence/similarity problems in 2-D figures	M3, M4, M6
MA.912.GR.2.1	Describe/represent a transformation with coordinates	M10
MA.912.GR.2.2	Identify distance-preserving transformations	M10
MA.912.GR.2.3	Sequence of transformations mapping a figure	M10
MA.912.GR.2.4	Symmetries of reflection/rotation/translation	M10
MA.912.GR.2.5	Draw a transformed figure on the plane	M10
MA.912.GR.2.6	Rigid transformations to justify congruence	M3, M10
MA.912.GR.2.7	Justify congruence criteria via rigid transformations	M3
MA.912.GR.2.8	Transformation to justify similarity	M4, M10
MA.912.GR.2.9	Justify similarity criteria (non-rigid transformations)	M4
MA.912.GR.3.1	Weighted average of points on a line (partitioning)	M9
MA.912.GR.3.2	Coordinate geometry to classify/justify figures	M9
MA.912.GR.3.3	Coordinate-geometry problems (lines/circles/triangles/quads)	M9
MA.912.GR.3.4	Coordinate geometry: perimeter/area of polygons	M9
MA.912.GR.4.1	Cross-sections of 3-D figures	M11
MA.912.GR.4.2	3-D objects from rotations of 2-D figures	M11
MA.912.GR.4.3	Dilations and their effect on area/SA/volume	M8
MA.912.GR.4.4	Area of 2-D figures	M8
MA.912.GR.4.5	Volume of 3-D figures	M8
MA.912.GR.4.6	Surface area of 3-D figures	M8
MA.912.GR.5.1	Construct a copy of a segment or angle	M1

Standard	Description (abbreviated)	Taught in
MA.912.GR.5.2	Construct bisectors, incl. perpendicular bisector	M1 (M2 proofs)
MA.912.GR.5.3	Construct inscribed/circumscribed circles of a triangle	M7
MA.912.GR.5.4	Construct a regular polygon inscribed in a circle	M7 (honors)
MA.912.GR.5.5	Construct a tangent from an external point	M7 (honors)
MA.912.GR.6.1	Length of secant/tangent/segment/chord	M7
MA.912.GR.6.2	Measures of arcs and related angles	M7
MA.912.GR.6.3	Triangles/quadrilaterals inscribed in a circle	M7
MA.912.GR.6.4	Arc length and area of a sector	M8
MA.912.GR.6.5	Prove all circles similar via transformations	M7
MA.912.GR.7.2	Derive/create the equation of a circle	M11 (M9 coordinate)
MA.912.GR.7.3	Graph/solve problems with a circle equation	M11
MA.912.LT.4.3	If-then / iff / all / not; converse, inverse, contrapositive	M2
MA.912.LT.4.8	Construct proofs, including proof by contradiction	M2 (applied throughout)
MA.912.LT.4.10	Judge validity of arguments; give counterexamples	M2
MA.912.T.1.1	Define trig ratios for acute angles	M5
MA.912.T.1.2	Right-triangle problems with trig ratios + Pythagorean	M5
MA.912.T.1.3	Law of Sines and Law of Cosines	M5 (honors)
MA.912.T.1.4	Area of a triangle from two sides and included angle	M5

[VERIFY] module placement. The map above is derived from lesson topics and the CPALMS course description, not from an explicit per-lesson alignment file (none is stored in the build). The pairings are sound, but confirm any borderline case (e.g., GR.6.4 arc length/sector taught in M8 vs M7; GR.7.2 in M11 vs M9) against your section's scope-and-sequence before reporting alignment formally.

12 What makes this course classical & cultivating culture

Classical moves this course actually makes: it treats geometry as Euclid did — a discipline of *proof*, not just calculation. Scholars construct with compass and straightedge (the oldest mathematical tools), reason from postulates to theorems, and in the honors strand prove by contradiction and derive the Laws of Sines and Cosines. The Commonplace Corner on the course home carries a line worth keeping each week. Wonder comes first (why must these be true?), mechanism second.

Cultivating culture asynchronously: name good reasoning publicly when you see it; when a scholar finds a valid alternate proof, celebrate it in the weekly announcement. Model curiosity — pose the week's "why" before the "how." Hold the virtue thread visible: perseverance is the geometry virtue, and a hard proof is the place to practice it. Treat feedback as relationship, not correction.

13 Sensitive-content notes

Geometry Honors carries **no mature, violent, or contested content** requiring an advance family script. The only advisories are practical:

- **VR comfort & safety.** Some scholars experience motion discomfort in VR. Offer the non-VR alternative (per the syllabus) without penalty, and remind scholars to clear a safe physical space.
- **Physical tools.** Constructions use a compass with a metal point — a routine safety note for younger scholars in a section.

14 Adapting the asynchronous course to a live class (2 days/week)

The On-Demand build is the source of truth; a live section teaches the *same* lessons, assignments, and assessments, just on a shared schedule. With **two meetings a week**, the guiding principle is: **use live time for the things that are hard to do alone** — modeling constructions, working proofs together, and resolving misconceptions — and push solo practice to between-class work.

The repeatable weekly rhythm

When	Do this live / async	Purpose
Day 1 (live, ~50 min)	Open the week's Essential Question. Teach/model the first 1–2 lessons; **demonstrate the construction or proof live** using the same lesson pages projected in VR or Teams. End by assigning the matching Independent Practice.	Front-load the hardest reasoning while you're in the room
Between classes (async)	Scholars complete the Independent Practice (2 attempts) and the paired VR resource on their own.	Solo practice with immediate auto-feedback
Day 2 (live, ~50 min)	Warm up on the 1–2 items most scholars missed (pull from practice analytics). Teach the remaining lesson(s). Run the **Mid-Module Check or a construction/proof workshop** as guided practice, circulating for feedback.	Close gaps; do teacher-graded work with support
End of week (async)	Scholars submit any construction/proof and, at module's end, the Summative Review (completion) then the Summative Assessment.	Independent demonstration of mastery

Weekly suggestion, module by module

At two 50-minute meetings a week, plan roughly **2–3 weeks per module** (11 modules across ~34 teaching weeks). Each week, adapt as follows:

- **Lessons → live mini-lecture.** Convert each week's lesson page into a 15–20 minute worked walk-through. The lesson's worked examples become your live examples; the "reveal" interactives become your call-and-response questions.
- **Independent Practice → homework.** Keep it exactly as built; it is the between-class assignment and your Day-2 warm-up data.
- **Constructions & proofs → live workshop.** These are the highest-value live minutes — model one, then have scholars attempt one while you circulate. This is where a live section most outperforms self-pacing.
- **VR resource → shared or solo.** If your meeting is in VR, do the resource together as the immersive anchor; otherwise assign it between classes.
- **Mid-Module Check → live checkpoint.** Use it as an in-class formative; reteach on the spot before scholars go on to the summative.
- **Summative Review & Assessment → async, proctored as your school directs.** Keep the review for completion and the assessment for correctness, unchanged.

> **Heavy weeks to protect (live).** Give Modules **1** and **7** (construction-heavy) and Modules **2, 3, 6** (proof-heavy) an extra live workshop day or a longer Day-2 if your schedule allows — these are where live instruction pays off most, and they line up with the workload spikes in §10.

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